Measurement-Calibrated Procedural Generation of Boiling Foam

A generative algorithm fitted to video measurement

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Abstract

I present a generative algorithm for the foam on a boiling liquid surface, in which every operator and every constant is fitted to statistics measured from a single handheld video. The algorithm maintains a population of surface cells under seven operators: nucleation, advection, contact relaxation, condensation, coalescence, destruction and capillary oscillation. A pinhole camera model projects each cell through its local Jacobian, so perspective magnification and foreshortening emerge from geometry rather than from tuned constants.

The methodological contribution is negative. Three mechanisms can produce the spatial size gradient the footage shows, and two of them are falsified by the same measurements used to calibrate the model. Coalescence-driven coarsening predicts that old foam is coarser and broader in distribution; the footage shows it is finer and narrower, with the log-radius spread falling from 0.65 to 0.40. Advective sorting reproduces the gradient visually but requires a surface current of 0.08 m/s, against a measured 0.011 m/s. The surviving mechanism is entirely local: birth radius scales with substrate superheat, and a churn term accelerates film drainage over the burner.

The generated sequence, measured by the unmodified analysis pipeline, matches the source on cell diameter (8.7 to 36.9 px against 9.1 to 42.7), median intensity (110 against 114), intensity spread (35 against 36) and three-frame survival (0.48 against 0.51).

1. Introduction

A boiling surface is a hard generative target because it has no persistent structure. Nothing in the image survives long enough to be animated: the measured mean cell lifetime is 0.141 s, and 17.5% of pixels change by more than 30 grey levels in every frame. What persists is a distribution, so the object to be generated is a population rather than a shape.

The usual graphics approach is a particle system with hand-tuned spawn rates, sizes and lifetimes, adjusted until a frame looks right [10]. This fails in a specific and instructive way. A single still frame constrains the size distribution but says almost nothing about the processes that maintain it, and several mutually incompatible process models produce indistinguishable stills. I encountered exactly this: an advection-based model reproduced every static feature of the footage and was wrong by a factor of seven in surface velocity.

This paper takes the opposite route. We first extract a set of target statistics from the source video, then construct operators whose parameters are pinned to those statistics, and finally re-run the identical measurement pipeline on the generated output.

Contributions.

1. A measurement protocol that converts an uncalibrated handheld video into target statistics for a generative model, covering size, spatial gradient, motion, lifetime and appearance.
2. A seven-operator generation algorithm for surface foam, running at interactive rates for populations of about 3,000 cells.

3. Falsification of two standard mechanisms for the observed spatial gradient, each killed by a specific measurement, and identification of a purely local mechanism that survives.
4. Two rendering results that affected realism more than any physical parameter: drawing cells largest-first rather than depth-sorted, and damping the capillary oscillation that otherwise aliases against the frame rate.

2. Related work

Three literatures bear on this problem, and none of them answers it directly.

Foam physics. Dry foam structure is governed by Plateau's laws and minimal-surface arguments [1], with Ostwald ripening driving slow coarsening as gas diffuses from small cells to large ones [2]. Recent work characterises the resulting bubble-size distribution in the wet limit and its approach to a scaling state [13, 14]. This literature describes foams that are near equilibrium and evolve over minutes. A rolling boil is neither: cells here live 0.14 s and the population is renewed several times a second. Ripening is far too slow to matter, which is the first reason coalescence-style coarsening was a poor fit.

Packed-disc dynamics. Discrete element methods [6] and position-based dynamics [5] both resolve contacts in dense particle assemblies, the latter by correcting positions directly rather than integrating contact forces. Position-based relaxation is the right tool for jammed foam because it is unconditionally stable at large timesteps and needs no stiffness tuning. We use it unchanged and claim no contribution there.

Procedural generation in graphics. Particle systems [10] and procedural textures can produce plausible foam, and the standard workflow tunes parameters against a reference image by eye. Related graphics work generates diffuse material from an underlying fluid solve, either as classified spray, foam and bubble particles [11] or as an explicit weighted-Voronoi foam geometry [12]; both take the flow field as given rather than calibrating a generator against footage. Our objection is not that this looks bad but that it is underdetermined: a still frame does not distinguish process models, and the discriminating information lives in the temporal statistics.

Single-bubble physics. Two results are used directly. Diffusion-limited dissolution of a gas bubble [3] gives a shrinkage law whose form we adapt to a surface cap, where flux scales with exposed area while volume scales with radius cubed. Rayleigh's analysis of capillary oscillation [4] gives the mode-2 frequency used for the shimmer term.

The gap this work addresses is calibration. Foam physics gives equilibrium structure, packed-disc methods give contact resolution, and graphics gives rendering, but none supplies a procedure for deciding which generative mechanism a given piece of footage actually supports.

3. Target statistics from video

The source is a single handheld clip: 478 × 850 px, 30 fps, 23.5 s, a pan viewed obliquely from above with the burner visible at the near rim. No calibration target, no known pan size, no controlled lighting. Six measurements are extracted.

Characteristic cell size. Individual cells in a jammed foam resist segmentation, so size is taken from the radial power spectrum of 128 × 128 patches rather than from detected objects. The dominant wavelength gives cell diameter as a function of image row, robust to the fact that cells touch and deform.

Perspective squash. One-dimensional autocorrelation along rows and columns gives the structure period in each direction. Their ratio is the projected axis ratio q.

Size distribution. Laplacian-of-Gaussian blob detection [8] on the intensity-inverted frame locates dark cell cores; a log-normal is fitted per depth band. The fitted spread σ, not the median, carries the discriminating

information.

Motion. Farnebäck optical flow [7] is computed per frame pair. The mean of the per-frame speed and the speed of the time-averaged field are reported separately; their ratio separates turbulent jitter from bulk current.

Lifetime. Intensity autocorrelation against lag gives a decorrelation timescale. A nearest-neighbour tracker over detected cores gives an explicit survival curve, which is the stronger constraint.

Appearance. Intensity percentiles, mean channel values, and the mean radial intensity profile averaged over roughly 1,000 detected cells.

Quantity	Measured value
Cell diameter, far rim	9.1 px
Cell diameter, near rim	42.7 px
Projected axis ratio q	0.83 to 0.94
σ, near burner	0.65
σ, far from burner	0.40
Instantaneous speed	1.43 px/frame (43 px/s)
Time-averaged drift	0.41 px/frame
Jitter fraction of motion	71%
Mean tracked lifetime	0.141 s
Lifetime p90	0.333 s
Survival at 3 / 5 frames	0.51 / 0.31
Intensity p1 / median / p99	51 / 114 / 223
Intensity std	36
Mean BGR	118 / 116 / 122
Radial profile peak	see section 8.2
Pixels changing >30 levels per frame	17.5%

Table 1. Target statistics measured from the source clip.

Two entries do the heavy lifting later. The spread narrowing away from the burner, from 0.65 to 0.40, rules out coarsening. The drift of 0.41 px/frame, about 0.011 m/s at the pan scale, rules out advective sorting.

4. State representation

The foam is a set of N surface cells on the plane, inside a disc of radius R_{pan}. All quantities are SI. The cell centre lies in pan coordinates with one axis pointing away from the camera, so the burner offset and the image row gradient share that axis.

Each cell carries eight attributes, held as parallel arrays rather than objects:

Attribute	Type	Role
position	2-vector, m	centre on the liquid plane
radius	scalar, m	cap radius, the only size state
age	scalar, s	time since nucleation; drives drainage
turbulent velocity	2-vector, m/s	Ornstein-Uhlenbeck state, persists across steps
oscillation phase	scalar, rad	fixed at birth, randomises the wobble

Attribute	Type	Role
deformation axis	scalar, rad	fixed at birth, orients the mode-2 ellipse
shading jitter	scalar	per-cell brightness multiplier, 7.5% spread
generation	integer	coalescence count, diagnostic only

Table 2. Per-cell state. Eight attributes, stored as parallel arrays.

The layout is structure-of-arrays because every operator is a whole-population vector operation. Nothing iterates over cells in the physics; the only per-cell loop in the system is the rasteriser. Birth appends to each array and death compacts with a boolean mask, so N varies freely between steps and no capacity is reserved beyond a safety ceiling of 9,000.

Two attributes deserve comment. Age is stored rather than derived because two separate operators consume it, drainage and oscillation damping, with different time constants. Radius is the sole size state: there is no separate volume or film-thickness variable, and cap volume is reconstructed as needed on the assumption of a hemispherical cap.

Neighbour queries are served by a k-d tree rebuilt once per substep. The same query result feeds contact relaxation, coalescence and the crowding test, so the tree cost is amortised across three operators.

5. Generation operators

Seven operators act on the population each substep. Each is stated with its law, the measurement that pins it, and the reasoning behind its form.

5.1 Nucleation

Births are Poisson with rate λ per second over the whole pan. The operator’s one structural decision is that position and size are drawn independently.

Position is a mixture: a fraction 0.40 from a Gaussian of width 0.062 m centred on the burner at x_h, the remainder uniform over the pan disc. The whole base nucleates; the burner only concentrates it.

Size is log-normal, scaled by the local superheat:

$$r_0 = exp(N(\ln r_{med} , \sigma_b)) \cdot [\varepsilon + (1 - \varepsilon) \cdot exp(- ||x - x_h||^2 / 2\sigma_r^2)] \quad (1)$$

with median radius r_{med} = 8.80 mm, σ_b fitted to the measured near-burner spread, and edge fraction ε = 0.40. A cell leaving the base over the flame is large; one leaving the cooler rim is 40% of that. This decoupling is what lets the far field stay fine without any transport.

A birth is rejected if it lands within 0.42 of an existing cell’s radius. A cell can only surface where there is free liquid, so packing becomes self-limiting rather than a tuned target.

5.2 Advection

Velocity is a weak mean flow plus a turbulent fluctuation. The mean field is the surface signature of a convection roll: radial outflow from the plume with a tanh ramp and exponential decay, a 12% tangential swirl, and a sweep of 0.0062 m/s toward the far rim, all braked by 85% as cells pack against the wall.

The fluctuation is an Ornstein-Uhlenbeck process [9], updated exactly rather than by Euler steps:

$$u' = a \cdot u + (1 - a^2)^{1/2} \cdot \sigma_u \cdot \xi, \quad a = exp(-dt / \tau) \quad (2)$$

with σ_u = 0.0070 m/s and τ = 0.130 s. The exact update keeps the stationary variance independent of timestep. OU rather than white noise matters because white noise decorrelates every frame and produces visible

strobing; a correlation time of 0.13 s gives motion that persists for about four frames.

The split between mean flow and fluctuation is set directly by the measurement: 71% of observed motion is jitter, so σ_u dominates the sweep by roughly a factor of two.

5.3 Contact relaxation

Overlaps are resolved by position-based correction [5] against the seven nearest neighbours, three passes per substep. For an overlapping pair the correction splits by inverse area, so large cells move less and shoulder fine foam aside, with stiffness k = 0.42. Positions are corrected directly, so there is no force integration and no stability limit on the timestep. The antecedent for the packing treatment is the discrete element method [6].

5.4 Condensation

A steam cap at a free surface loses volume as vapour condenses through its exposed area. Flux scales with cap area, as r^2, while cap volume scales as r^3, so the radius falls at a constant rate; the form is adapted from diffusion-limited bubble dissolution [3]. Shrinking stalls when the Laplace overpressure of a small cap balances the driving term:

$$dr/dt = -c \cdot max(0 , 1 - r_{eq}^{\,2} / r^2) \quad (3)$$

with r_{eq} = 1.31 mm. This operator is the fining mechanism. Because every surviving cell is drawn toward one equilibrium radius, an initially broad distribution becomes narrow with age, which is precisely the measured fall in σ from 0.65 to 0.40. No other operator in the system narrows a distribution.

5.5 Coalescence

Touching neighbours merge at a low rate, conserving gas volume:

$$r_{merged} = (r_i^3 + r_j^3)^{1/3} \quad (4)$$

The pair qualifies when separation is within 3% of contact, and the rate is 0.18 per second, rising modestly with the age of the younger partner. A cell may take part in at most one merge per step, enforced by three vectorised passes over the candidate list rather than a sequential greedy loop.

This operator is deliberately weak. It was the original candidate for the spatial gradient and is retained only because film rupture between touching cells is real, not because the footage demands it.

5.6 Destruction

Cells are removed by a hazard rate with three factors:

$$h = h_0 \cdot (r / r_{ref})^{3.2} \cdot (1 + churn(x) \cdot age / t_{drain}) \quad (5)$$

with h_0 = 0.26 per second, r_{ref} = 2.06 mm, t_{drain} = 2.2 s and churn gain 45. Cells are also removed outside the range 1.03 mm to 23.0 mm.

The size exponent of 3.2 makes the hazard strongly selective: a cell twice the reference radius is roughly nine times more fragile. The churn factor is the second half of the surviving mechanism. Over the burner, rising vapour continuously tears the raft open, so film drainage is accelerated forty-fivefold and no persistent foam accumulates. Away from it the fine raft survives for seconds and packs solid.

Critically, churn multiplies the age term rather than the whole hazard. An earlier formulation multiplying the entire rate destroyed freshly nucleated cells over the burner as well, collapsing the population to a quarter of its target. Fresh cells should burst on their own schedule; only aged films are torn.

5.7 Capillary oscillation

Each cell carries a Rayleigh mode-2 oscillation [4] at

$$f = (1 / 2\pi) \cdot [8\sigma / (\rho\, r^3)]^{1/2} \quad (6)$$

with σ = 0.059 N/m and ρ = 958 kg/m^3, values for water at 100°C. Amplitude is 3.0% at birth, scaled mildly by radius, and damped viscously with a time constant of 0.075 s.

The damping is not cosmetic. For a 2 mm cell the frequency is about 39 Hz, which at 30 fps advances the phase by 8.3 rad per frame: undamped, every cell is in an uncorrelated phase each frame and the foam strobes. Damping confines full-amplitude ringing to the first few frames after a cell surfaces, turning a flicker into a shimmer.

6. The step algorithm

One substep applies the seven operators in a fixed order. Six substeps run per rendered frame, giving dt = 5.6 ms at 30 fps.

```
STEP(state, dt):
  1  NUCLEATE
       k ~ Poisson(lambda * dt)
       draw k positions: 40% Gaussian at burner, 60% uniform on disc
       draw k radii:     log-normal, scaled by local superheat
       reject births landing inside 0.42 * r of a nearest neighbour
       append survivors

  2  if N < 3: return

  3  ADVECT
       u <- a*u + sqrt(1-a^2) * sigma_u * randn     (OU, a = exp(-dt/tau))
       v <- mean_flow(x) + u
       x <- x + v*dt
       clamp to pan wall, reflect turbulent component

  4  NEIGHBOURS
       build k-d tree; query 7 nearest for every cell    <- shared by 5,6

  5  RELAX                                              (3 passes)
       for each overlapping pair: split correction by inverse area

  6  CONDENSE
       r <- r - c * max(0, 1 - r_eq^2/r^2) * dt

  7  COALESCE
       candidates: pairs within 3% of contact, i < j
       accept with probability 1 - exp(-rate(age) * dt)
       de-conflict: permute, unique on i, unique on j, drop chains
       merge: r <- (r_i^3 + r_j^3)^(1/3), position by volume weight

  8  age <- age + dt

  9  BURST
       h <- h_0 * (r/r_ref)^3.2 * (1 + churn(x) * age/t_drain)
       kill where U(0,1) < 1 - exp(-h*dt), or r out of [r_min, r_max]
       compact arrays
```

Algorithm 1. One substep. Operators run in the order shown; the ordering constraints are given below.

Why this order. Three constraints fix it. Relaxation must precede coalescence, because the contact test is a geometric predicate and reading it before overlaps are resolved would merge cells that are merely interpenetrating rather than touching. Condensation must precede coalescence for the same reason: it changes radii and therefore contact. Ageing is incremented after coalescence but before bursting, so a merged cell inherits the younger partner’s age and is not immediately destroyed by a drainage term it did not earn.

Substepping. Six substeps per frame is set by the OU correlation time, not by stability. Position-based relaxation is stable at any timestep, but $\tau = 0.130$ s must be resolved by at least a few steps or the turbulent velocity loses its correlation structure. Fewer than four substeps visibly flattens the motion.

Complexity. The tree build and query dominate at $O(N \log N)$; every other operator is $O(N)$ or $O(Nk)$ with k fixed. Coalescence de-confliction is three sorts, $O(m \log m)$ in the candidate count. Nothing is quadratic and nothing loops over cells in Python.

Measured cost. At N around 3,000: 10 to 14 ms per substep, so roughly 70 ms of physics per rendered frame, against 200 to 260 ms for the rasteriser. Rendering is the bottleneck by a factor of three, which is why the sprite lookup table of section 8 matters more for throughput than any physics optimisation.

Steady state. Population is set by the balance of birth against hazard and is not prescribed. With the parameters of section 10 the system settles near 2,800 cells within about 1.5 s of simulated time, from an empty pan. The spin-up routine runs 2 s before the first frame is rendered. Area coverage settles above unity because the raft is multi-layered in projection; the crowding rejection in nucleation, not a coverage target, is what stops it running away.

7. Mechanism selection by falsification

The footage shows a clear spatial gradient: fine, dense, nearly monodisperse foam far from the burner, and coarse, sparse, broadly distributed cells near it. Three mechanisms can produce this. Two are ruled out by the measurements.

7.1 Coalescence-driven coarsening

Hypothesis. Cells merge on contact; foam that has survived longer has merged more, so the population coarsens with age. A spatial gradient follows if age varies spatially.

Falsified by σ. Coalescence is a broadening process. Merging samples from a distribution and summing volumes increases the spread of the result, and the same is true of the diffusive route to coarsening, where the size distribution broadens as it approaches a scaling state [2, 13, 14]. Older foam should therefore be both coarser and more variable. The measurement shows the opposite: σ falls from 0.65 near the burner to 0.40 away from it. The distribution narrows exactly where the hypothesis requires it to widen.

This is a sign error, not a magnitude error, so no parameter choice rescues it. A model built on coarsening reached σ of 0.35 uniformly across the pan with no gradient at all.

7.2 Advective sorting

Hypothesis. Cells nucleate coarse over the burner, and a surface current carries them away. Size-selective bursting strips the large cells in transit, so only fine foam arrives at the far rim. This explains the gradient and the narrowing together, since a truncated tail is a narrower tail.

Provisionally accepted, then falsified by optical flow. This model reproduced the size profile well, matching the far-rim diameter exactly and the near rim within 20%. It also reproduced the narrowing. On static evidence it was correct.

The requirement is transport: a cell must cross a substantial fraction of the pan within its lifetime. With a mean lifetime of 0.141 s and a transport distance of order 0.10 m, the implied current is around 0.08 m/s. Running the measurement pipeline on the generated video returned 7.62 px/frame against 1.43 measured, and a drift fraction of 39% against 29%.

The surface in the footage barely flows. Any mechanism requiring bulk transport within a cell lifetime is excluded by a factor of roughly seven, and reducing the current to the measured value collapsed the gradient entirely: near-field diameter fell from 42 px to 15 px, matching the far field.

7.3 Local superheat with churn

Hypothesis. No transport. Two position-dependent terms acting where the cell already is.

Birth radius scales with substrate superheat, so vapour leaves the base largest over the flame and smallest at the cool rim. Independently, the burst hazard gains a churn factor over the burner, where rising vapour tears open any raft that tries to form. Away from the burner, fine cells relax toward the condensation equilibrium radius and persist for seconds, packing solid.

Consistent with all six measurements. The gradient arises without transport, so it survives the measured surface velocity. The narrowing arises from condensation drawing survivors toward a single equilibrium radius, which is a contracting map on the distribution. The lifetime distribution is reproduced by the hazard model. Nothing in the mechanism requires a cell to move more than its own diameter.

7.4 The methodological point

The advection model passed every check available from a still frame and was wrong by a factor of seven in a quantity a still frame cannot show. Spatial statistics did not discriminate between it and the correct model; a temporal statistic did, and only because the measurement pipeline was built as a reusable artefact and run unchanged on the generated output.

This suggests a general discipline for calibrated procedural generation. Fit against a static reference and several incompatible process models will fit equally well. The discriminating evidence is in the temporal statistics, and it is only accessible if validation is automated rather than visual.

8. Rendering algorithm

Rendering is a purpose-built rasteriser rather than a general vector backend. Two decisions carry most of the realism, and neither is a physical parameter.

8.1 Projection through the local Jacobian

A pinhole camera sits 0.185 m from the pan centre at 52 degrees elevation. Rather than projecting a circle and fitting an ellipse, we project the centre and take the 2 × 2 Jacobian J of the world-to-image map there.

A unit world circle of radius r maps to the ellipse traced by J, so inverting J and normalising by r gives a scalar that is 1 exactly on the cell boundary. The mode-2 deformation matrix from section 5.7 is composed into J before inversion.

This is exact to first order and costs four finite differences per cell. Both the perspective size gradient and the measured axis ratio q fall out of it: the model reproduces q between 0.83 and 0.94 without any flattening constant. An earlier version that scaled radii by depth and squashed by a fixed factor needed two tuned constants and still misplaced the gradient.

8.2 Sprite lookup table

The measured radial profile of section 3 is baked once into a 224 × 224 table over normalised cell coordinates. It contains the dark core rising as a power of normalised radius, a Gaussian rim peaking near the boundary with width 0.050, a tight specular catch-light on the lit side, a weak bounce term on the shadow side, and a dark Plateau border fading out beyond the rim.

Each cell then costs a coordinate transform, an integer gather and an alpha blend, with no transcendental functions in the inner loop. This is what makes 3,000 cells per frame affordable in array code.

8.3 Draw order

Cells are drawn largest first, smallest last.

The obvious choice is a painter's algorithm sorted by depth, and it is wrong. Foam cells are coplanar: they share Plateau borders rather than occluding one another [1]. Depth sorting makes each cell's lower arc get overpainted by its nearer neighbour, leaving only bright upper crescents. The result is a convincing fish-scale texture that is not foam, and it survived several rounds of visual inspection before being identified.

Sorting by radius instead reproduces what a real foam does: large cells tile the surface and small ones nestle into the interstices, so complete rings are visible. Changing only the sort key moved the intensity spread from 38 to 29 against a measured 31.

8.4 Temporal treatment

The capillary oscillation of section 5.7 aliases badly at 30 fps and is damped for that reason. Beyond it the renderer adds a moving ripple field on exposed liquid, a Gaussian blur of 0.75 px standing in for lens and compression softness, per-pixel grain at 3.0 levels, and a mild vignette. These bring the intensity spread and percentile structure into agreement with the source but carry no physical content.

9. Results and validation

The generated sequence is 10 s at 30 fps, 478 × 850 px, rendered from a steady state of about 2,800 cells. Validation runs the analysis script of section 3 unmodified on the generated file. Nothing in the pipeline is aware of which video it is reading.

Quantity	Filmed	Generated
Cell diameter, far rim	9.1 px	8.7 px
Cell diameter, near rim	42.7 px	36.9 px
σ, near burner	0.65	0.56
σ, far from burner	0.40	0.38
Intensity p1	51	72
Intensity median	114	110
Intensity p99	223	220
Intensity std	36	35
Mean lifetime	0.141 s	0.115 s
Lifetime p90	0.333 s	0.233 s
Survival at 3 frames	0.51	0.48
Survival at 5 frames	0.31	0.25
Radial profile peak	same normalised radius	same normalised radius

Table 3. Source clip against generated sequence, both measured by the same unmodified pipeline.

Appearance agreement is close. Median intensity is within 4 levels, the spread within 1, and the radial profile peaks at the same normalised radius, which is the direct test of the sprite model.

The size gradient is reproduced in shape but compressed at the near end: generated near-rim cells are about 14% under target while the far rim matches within 5%. The cause is the interaction between birth radius and the hazard exponent. Raising the birth median to close the near-field gap shortens large-cell lifetimes as the cube of radius, so the cells vanish before they are seen, and the population thins visibly. The two constraints are in tension and the present values are a compromise.

The lifetime distribution is slightly short throughout. Both measurements share a tracker whose failures bias the estimate downward, so the absolute values are suspect in the same direction for both videos; the survival

ratios are the more trustworthy comparison, and those agree within 0.06.

The residual on p1 is the clearest defect: generated dark cores reach 72 where the source reaches 51. The source has genuinely darker interstices than the model produces, discussed in section 11.

10. Parameters

All values as used in the reported run. “Pinned” means the value is fixed by a measurement or a physical constant; “fitted” means it was chosen to bring a measured output into agreement; “free” means it affects appearance but no measured quantity.

Parameter	Value	Status	Source
Pan radius	0.105 m	free	assumed; trades off against focal length
Surface tension	0.059 N/m	pinned	water at 100°C
Density	958 kg/m^3	pinned	water at 100°C
Birth rate	17,000 /s	fitted	cell count and coverage
Birth radius median	8.80 mm	fitted	near-rim cell diameter
Birth radius sigma	0.46	pinned	measured σ near burner
Edge size fraction	0.40	fitted	far-rim cell diameter
Burner offset	(0.006, -0.050) m	fitted	position of the coarse region
Hot zone width	0.062 m	fitted	width of the coarse region
Hot birth fraction	0.40	fitted	far-field raft density
Crowding block	0.42	free	self-limits packing
Sweep	0.0062 m/s	pinned	measured drift, 0.41 px/frame
Plume outflow	0.0085 m/s	pinned	measured drift
Turbulent sigma	0.0070 m/s	pinned	measured jitter fraction, 71%
Turbulent tau	0.130 s	pinned	intensity autocorrelation decay
Neighbours	7	free	contact resolution
Relax stiffness	0.42	free	convergence rate
Condensation rate	5.0×10^{-3} m/s	fitted	rate of fining with distance
Equilibrium radius	1.31 mm	fitted	far-field cell diameter
Coalesce rate	0.18 /s	free	weak, no measurement depends on it
Burst rate	0.26 /s	fitted	survival curve
Burst reference radius	2.06 mm	fitted	survival curve
Burst size exponent	3.2	fitted	size gradient steepness
Drain time	2.2 s	fitted	raft persistence
Churn gain	45	fitted	contrast between near and far halves
Wobble amplitude	0.030	free	shimmer strength
Wobble damping	0.075 s	fitted	removes 30 fps aliasing
Camera distance	0.185 m	fitted	near-to-far magnification ratio
Camera elevation	52 deg	pinned	measured axis ratio q
Substeps per frame	6	free	resolves turbulent tau

Table 4. Complete parameter set for the reported run.

Of 30 parameters, 9 are pinned directly by measurement or physics, 15 are fitted against a measured output, and 6 are free. The free parameters affect appearance only and could be replaced by any reasonable value without changing a reported result.

11. Limitations and future work

Single-clip calibration. Every parameter is fitted to one 23.5 s video of one pan on one burner. Nothing here demonstrates that the mechanism generalises across vessel size, heat input or liquid composition. The churn gain in particular is a single number standing in for an unmodelled process, and its value is not transferable.

Unknown scale. Pan radius is assumed, not measured, and trades off directly against focal length. Absolute cell sizes in millimetres are therefore uncertain by whatever factor the true pan differs from 0.21 m diameter. All reported agreement is in pixels, which is unaffected, but the physical parameters should be read as relative.

Near-field size deficit. Generated near-rim cells run about 14% under target, for the reason given in section 9. Resolving it properly needs the hazard model to separate film rupture from cap collapse, so that large fresh cells can be large without being fragile.

Dark cores too light. Generated p1 is 72 against 51. The source has darker interstices than the model produces, most likely because real foam is a multi-layer raft whose lower cells are shadowed. The renderer treats the foam as a single layer with a lit liquid beneath.

No wet sheen model. The most visible qualitative gap. Real footage shows irregular, mobile liquid films and bright wobbling highlights between cells, particularly near the burner. The model has only clean rings over a rippled background, and looks correspondingly tidy.

Monolayer assumption. Projected area coverage exceeds unity, which the model handles by drawing overlapping discs rather than by representing raft depth. Cells have no vertical position and cannot sit on one another.

No explicit Plateau borders. Cell boundaries are circular arcs with a dark rim, not the shared films and three-way junctions of real foam [1]. At the high packing of the far-field raft this is visible on close inspection.

Tracker bias. Both lifetime figures come from a nearest-neighbour tracker that loses cells on detection dropout, biasing lifetimes downward. Absolute values should be treated as lower bounds; only the comparison between the two videos is sound.

The most useful extension would be a second and third source clip at different heat inputs. If birth radius and churn gain both scale with a single heat parameter, the mechanism would be doing real explanatory work; if each clip needs its own fit, the model is a description rather than an explanation.

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Reproducibility

The simulation, the measurement pipeline, the generated sequence and a side-by-side comparison were produced alongside this paper. The measurement script runs on any video and prints every statistic in sections 3 and 9, so the validation in this paper can be repeated end to end on a fresh render.